REQUIREMENTS

Problem Statement #21 — EV Charging Station Reservation System

Name: K S Anagha
SRN: PES1UG24CS208
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Funtional Requirements:

Requirement ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR001	Funtional	The system shall match the driver vehicle connector type with available station chargers and reserve a 30-minute charging window	High	Pass: Incompatible connector plugs hidden from driver; Fail: Two vehicles booked for same socket concurrently.	Prevents incompatible pairings and double-booking of a single socket; core matching function of the reservation system.
FR002	Funtional	The system shall allow an EV driver to search for and filter nearby charging stations by location, connector type, and current availability	High	Pass: a search within a 5 km radius returns only stations with a compatible, available socket, displayed within 3 seconds of the query.	Discovery of a compatible, available station is the first step a driver must complete before any reservation can be made.
FR003	Funtional	The system shall calculate the estimated power rate for a requested time and display it to the driver before payment is confirmed.	Medium	Pass: the rate shown on screen matches the engine's output and is displayed before the "Confirm Payment" is enabled	Gives the driver transparent, accurate pricing so they can make decision before payment
FR004	Funtional	The system shall allow a driver to cancel a confirmed reservation up to 30min before the slot start time and immediately release the socket for other drivers.	High	Pass: a cancelled slot becomes bookable by another driver within 60seconds of the cancellation being confirmed.	Maximizes station utilization and gives drivers flexibility to change plans without penalty if cancelled in time.
FR005	Funtional	The system shall process the driver's payment through the connected payment gateway and generate a digital receipt upon a successful transaction.	High	Pass: a receipt containing a transaction ID is generated within 5 seconds of gateway approval and emailed to the driver	Completes the reservation transaction and gives the driver a record of payment.
NFR001	Non Funtional	The system shall release a socket and apply a no-show penalty if the driver fails to plug in within 10 minutes of the slot start time.	High	Pass: benchmarking tests confirm the release and penalty trigger within the target time and useful during peak load.	Prevents a no-show driver from blocking a socket indefinitely, protecting availability for other drivers.
NFR002	Non Funtional	The system shall support at least 500 concurrent reservation requests across all stations without average response times exceeding 3 seconds	High	Pass: a load test with 500 concurrent simulated users shows a best response time under 3 seconds with zero failed transactions.	Ensures the platform remains responsive city-wide during peak hours, when reservation demand increase